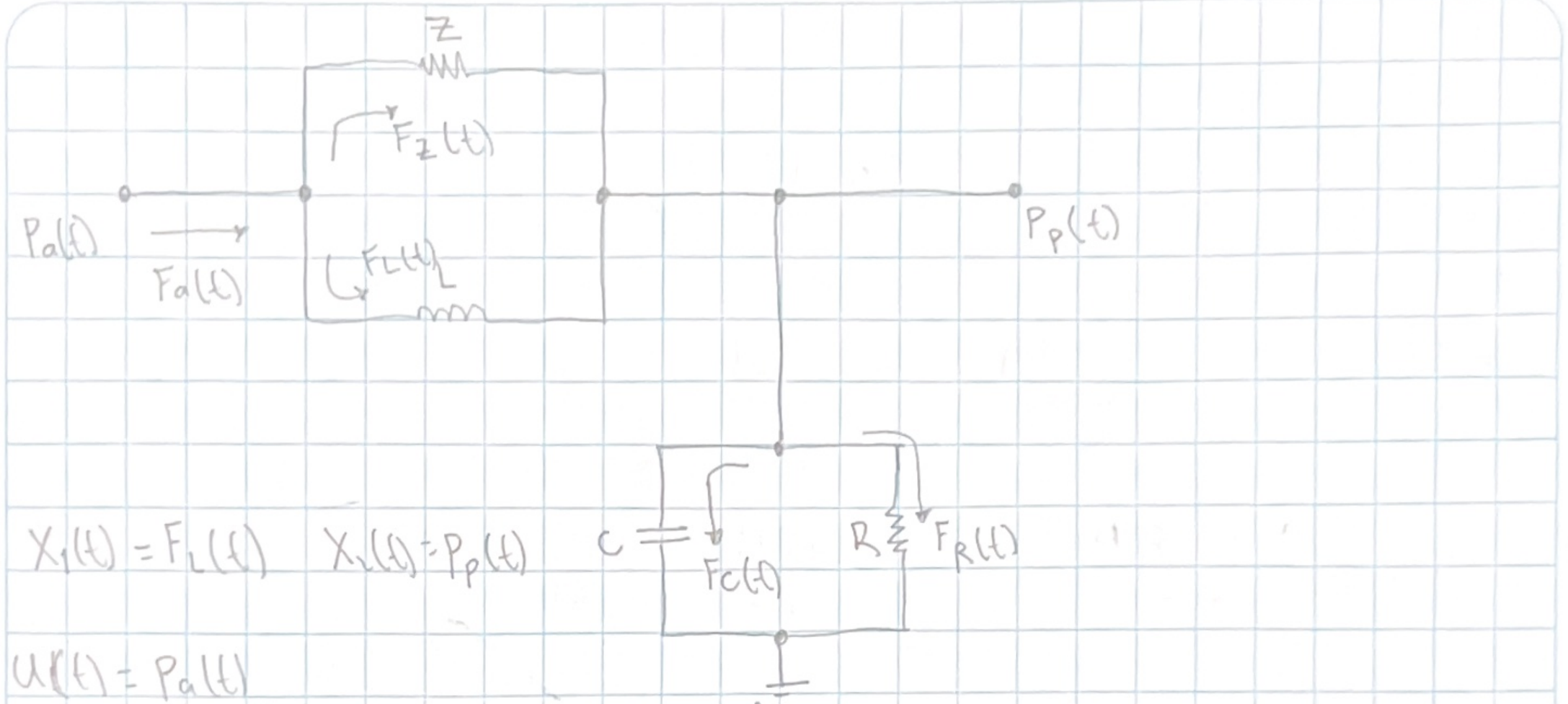
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Modelado de Sistemas Fisiologicos.



$X_1(t) = F_L(t)$ $X_2(t) = P_p(t)$

$U(t) = P_a(t)$

Para la corriente que pasa por el inductor...

$$P_a(t) - P_p(t) = L\frac{dF_L(t)}{dt} \rightarrow \frac{dF_L(t)}{dt} = \frac{P_a(t) - P_p(t)}{L}$$

Para la corriente que pasa por la resistencia...

$$P_a(t) - P_p(t) = Z F_Z(t) \rightarrow F_Z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

Para las corrientes que pasan por todos los componentes del circuito...

$$F_Z(t) + F_L(t) - F_C(t) - F_R(t) = 0 \rightarrow F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

donde F_R y F_C...

$$F_C(t) = C\frac{dP_p(t)}{dt} \qquad F_R(t) = \frac{P_p(t)}{R}$$

Sustituyendo todas las Relaciones se obtiene que... →

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Modelado de Sistemas Fisiologicos.

$$\frac{P_a(t) - P_p(t)}{Z} + F_L = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$C\frac{dP_p(t)}{dt} = \frac{P_a(t) - P_p(t)}{Z} + F_L - \frac{P_p(t)}{R}$$

$$\frac{dP_p(t)}{dt} = \frac{P_a(t) - P_p(t)}{ZC} + \frac{F_L}{C} - \frac{P_p(t)}{RC}$$

$$\frac{dP_p(t)}{dt} = \frac{P_a(t)}{ZC} + \frac{F_L}{C} - \left(\frac{1}{ZC} + \frac{1}{RC}\right)P_p(t)$$

Por lo tanto...

$$\frac{dP_p(t)}{dt} = \dot{P}_p(t) = \underbrace{\frac{P_a(t)}{ZC}}_{B_{21}} + \underbrace{\frac{F_L(t)}{C}}_{A_{21}} - \underbrace{\left(\frac{1}{ZC} + \frac{1}{RC}\right)}_{A_{22}}P_p(t)$$

$$\frac{dF_L(t)}{dt} = \dot{F}_L(t) = \underbrace{\frac{P_a(t)}{L}}_{B_{11}} - \underbrace{\frac{P_p(t)}{L}}_{A_{12}} + \underbrace{0 \cdot F_L(t)}_{A_{11}}$$

$$\dot{x}(t) = A \cdot x(t) + B u(t)$$

$$\dot{X}(t) = \begin{vmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\left(\frac{1}{ZC} + \frac{1}{RC}\right) \end{vmatrix} x(t) + \begin{vmatrix} \frac{1}{L} \\ \frac{1}{ZC} \end{vmatrix} u(t)$$

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Modelado de Sistemas Fisiologicos.

Considerando la presión sobre la distensibilidad arterial como salida...

$$y(t) = P_p(t) = [0 \quad 1]\, X(t)$$

Luego usando las siguientes ecuaciones...

$$F_L(t) + F_z(t) = F_C(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{z} \qquad F_C(t) = C\,\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R} \qquad F_L = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

Aplicamos Laplace...

$$F_L(s) + F_z(s) = F_C(s) + F_R(s) \qquad F_C(s) = C_s\, P_p(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{z} \qquad F_L(s) = \frac{1}{Ls}\left(P_a(s) - P_p(s)\right) \qquad F_R(s) = \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{Ls} + \frac{P_a(s) - P_p(s)}{z} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Ls} + \frac{P_a(s)}{z} = C_s P_p(s) + \frac{P_p(s)}{R} + \frac{P_p(s)}{z} + \frac{P_p(s)}{Ls}$$

$$P_a(s)\left(\frac{1}{Ls} + \frac{1}{z}\right) = P_p(s)\left(C_s + \frac{1}{R} + \frac{1}{z} + \frac{1}{Ls}\right)$$

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Modelado de Sistemas Fisiologicos.

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{1}{Ls} + \frac{1}{z}}{\frac{Cs}{1} + \frac{1}{R} + \frac{1}{z} + \frac{1}{Ls}} \qquad \frac{P_p(s)}{P_a(s)} = \frac{\frac{z + Ls}{Lzs}}{\frac{CRLzs^2 + Lzs + LRs + zR}{LRzs}}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{z + Ls}{Lzs}}{\frac{CRLzs^2 + L(z+R)s + zR}{LRzs}}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{CRs + 1}{R} + \frac{Ls + z}{Lzs}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{CRzs(z + Ls)}{Lzs(CRLzs^2 + L(z+R)s + zR)}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{zRs + LRs}{CRLzs^2 + L(z+R)s + zR}$$

Error Estacionario.

$$e(s) = \lim_{s \to 0} s \cdot Pa(s) \left(1 - \frac{P_p(s)}{Pa(s)}\right)$$

$$e(s) = \lim_{s \to 0} s \frac{1}{s} \left(1 - \frac{zR}{zR}\right) \qquad e(s) = \lim_{s \to 0} s \frac{1}{s}(0) \to \emptyset$$

$e(t) = 0\,V$

Estabilidad de los Polos.

$a = CLRz \qquad b = L(z+R) \qquad C = Rz$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hipotenso

$z = 0.010 \quad L = 0.001 \quad R = 1 \quad C = 3.500$

$a = 3.5 \times 10^{-5} \quad b = 1.01 \times 10^{-3} \quad C = 0.010$

$\lambda_1 = -14.429 + 8.8051i$

$\lambda_2 = -14.429 - 8.8051i$

Normotenso

$z = 0.010 \quad L = 0.001 \quad R = 1 \quad C = 1.500$

$a = 1.5 \times 10^{-5} \quad b = 1.01 \times 10^{-3} \quad C = 0.010$

$\lambda_1 = -12.062$

$\lambda_2 = -55.272$

Hipertenso

$z = 0.010 \quad L = 0.001 \quad R = 1 \quad C = 1.$

$a = 1 \times 10^{-5} \quad b = 1.01 \times 10^{-3} \quad C = 0.010$

$\lambda_1 = -10$

$\lambda_2 = -10$

Modelo de EFD

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int \left[P_a(t) - P_p(t)\right] dt = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$P_p(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \frac{P_a(t)}{Z} + \frac{1}{L}\int \left[P_a(t) - P_p(t)\right] dt - C\frac{dP_p(t)}{dt}$$

$$P_p(t)\left(\frac{Z+R}{ZR}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int \left[P_a(t) - P_p(t)\right] dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int \left[P_a(t) - P_p(t)\right] dt - C\frac{dP_p(t)}{dt}\right]\left(\frac{ZR}{Z+R}\right)$$